SCIENTIST GUIDE

Ratio Vitis 0.1.0 · Ratio Complexus

From field datum *to* *equation* and back

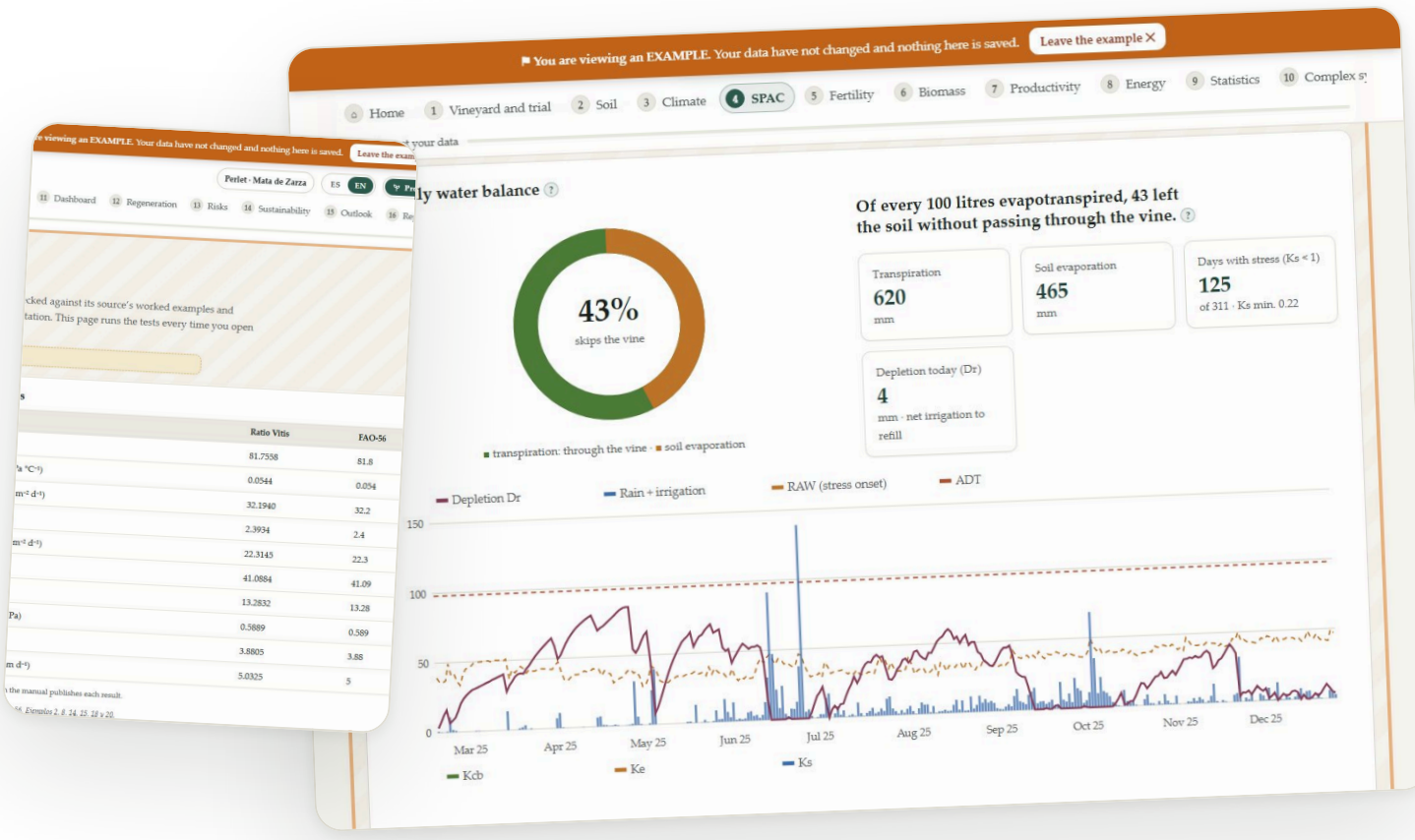
Model, equations, assumptions, validation and reproducibility of Ratio Vitis: the measured vineyard as a complex adaptive system.

1st edition · November 2026 · covers Ratio Vitis 0.1.0

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There is a companion guide for fieldwork: the Producer Guide.



1 Vineyard and trial
2 Soil
3 Climate
4 SPAC
5 Fertility
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8 Energy
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10 Complex system
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12 Regeneration
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FRONT MATTER

How to read this guide

Who it is for, what it documents and how to tell data, assumption and gap apart.

This guide is for whoever uses Ratio Vitis in research or teaching: the thesis student, the adviser, the committee, a journal reviewer. It documents each block's model, its equations with number and source, the flagged assumptions, how the engine was validated and how to reproduce every number outside the app.

Ratio Vitis is the app of objective 7 of Jorge Hernández-González's master's thesis: to assess grapevine adaptability in Mata de Zarza, Veracruz, as a complex adaptive system. The hypothesis is that fertilisation with organic manures that raise soil organic matter gives the vineyard more resilience at 1, 2 and 5 years (Hernández González, 2026).

Three kinds of datum

LABEL	MEANS	EXAMPLE
● **Published data**	Coefficient, threshold or equation taken from a cited source and checked against its text.	Ky = 0.85 (FAO-56, Table 24)
◆ **Ratio Vitis reading**	App assumption to complete a calculation; always flagged and editable.	RO = 0 and CR = 0 in the balance
no data	No measurement or source; nothing is computed from it.	Yield before harvest

? ARCHITECTURE

A single HTML file with its own CSS and JavaScript, no third-party libraries or resources; all computation runs in the browser. Sources are assembled with `armar.py` from `src/`. MIT licence.

METHOD

1 The vineyard as a system

Layers, flows and blocks: what goes into and comes out of each.

The vineyard is represented as a five-layer system —soil, plant, atmosphere, management and outcomes— joined by flows of water, energy, carbon and nutrients. Blocks 1 to 9 capture each layer's variables; Block 10 joins them in a signed directed network; blocks 11 to 16 read the system: dashboard, regeneration, risks, sustainability, outlook and report.

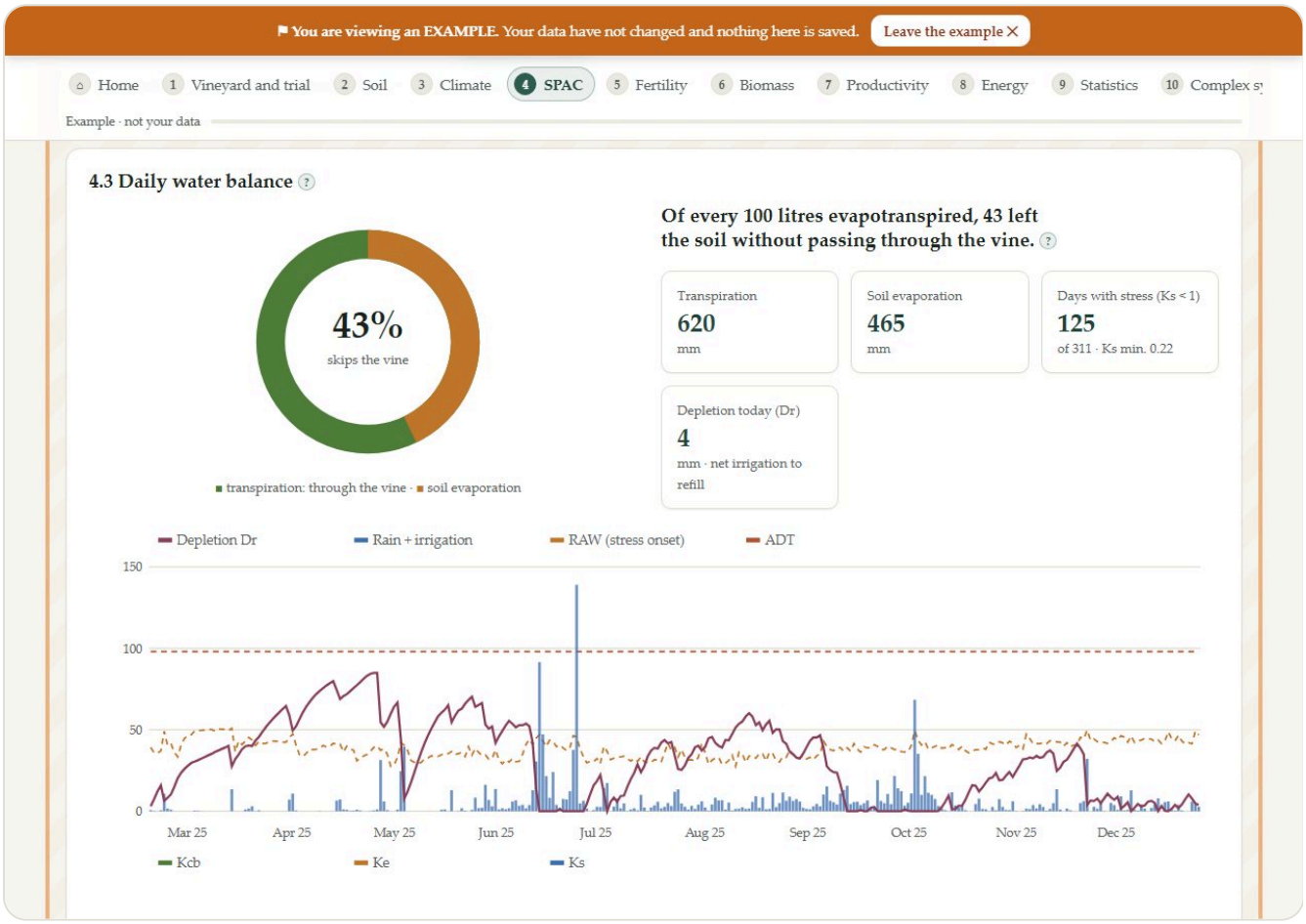


FIGURE 1.1 **The scientist view.** Each block closes with "The block model": equations, assumptions and sources.

BLOCK	INPUTS	MAIN OUTPUTS
1	INEGI locality or GPS; grape type; spacing; treatments, blocks and seed	Latitude, altitude, P, density, layout
2	Jar test, bulk density, θFC, θWP, Zr, pH, EC, OM	USDA class, TAW, RAW, salinity Yr
3	Daily CSV: T, RH, u2, Rs or n, P, Tdew	ET0, Winkler, GST, Huglin, CI, P/ET0
4	Stages, budburst, irrigations, ψ, phenology	Dr, Ks, Kcb, Ke, ETc adj, T, E, DP
5–7	Tissue and soil analyses, manures; pruning, LA, NDVI, DM; harvest	SR, DOP, dose; LAI, fc, C, CO_2, O_2; t ha^{-1}, Ravaz, WP, Ya/Ym
8–9	Climate and balance; field sheet	Rn, LE, H, β; ANOVA, Tukey, ANCOVA, repeated measures
10–16	All of the above	Network, loops, ISG, scenarios, report and JSON

METHOD

2 Equations by block

Each equation with its number in the source; they are the ones the engine runs.

2.1 Site and design

$$P = 101.3\ ((293 - 0.0065\ z) / 293)^{5.26}\ ;\ \gamma = 0.665 \times 10^{-3}\ P$$

FAO-56, Eq. 7–8

The randomised complete block draw shuffles treatments within each block with the mulberry32 generator seeded from a text seed; the same seed reproduces the layout. The field sheet exports one row per vine with block, plot and treatment.

2.2 Soil

$$TAW = 1000\ (\theta_{FC} - \theta_{WP})\ Z_r\ ;\ RAW = p \cdot TAW$$

FAO-56, Eq. 82–83

$$Y_r = 100 - b\ (EC_e - EC_{e,threshold})\ ;\ \text{vine: } 1.5\ \text{dS m}^{-1},\ b = 9.6$$

FAO-56, ch. 8

Texture class follows the USDA triangle. Without measured θFC and θWP the centre of the FAO-56 Table 19 range for the class is used, flagged as suggested. Bulk-density thresholds by texture are from USDA-NRCS; the 5.5–6.5 ideal pH, from NC State Extension.

2.3 Climate

$$ET_0 = [0.408\ \Delta\ (R_n - G) + \gamma\ (900/(T + 273))\ u_2\ (e_s - e_a)] / [\Delta + \gamma\ (1 + 0.34\ u_2)]$$

FAO-56, Eq. 6

With missing data the FAO-56 routes are followed: e_a from dew point (Eq. 14) or from RHmax and RHmin (Eq. 17); R_s from sunshine hours (Eq. 35) or Hargreaves with k_{Rs} = 0.16 inland and 0.19 on the coast (Eq. 50); wind from another height taken to 2 m (Eq. 47); without wind, 2 m s^{-1}. The app reports, day by day, which route it used.

$$W = \Sigma_{1\ Apr}^{31\ Oct} \max(0, T_{mean} - 10)\ ;\ GST = \text{mean } T_{mean} \text{ from 1 Apr to 31 Oct}$$

Amerine and Winkler (1944); del Río et al. (2023)

$$HI = \Sigma_{1\ Apr}^{30\ Sep} [(T_{mean} - 10) + (T_{max} - 10)] / 2 \cdot d\ ;\ CI = \text{mean September } T_{min}$$

Tonietto and Carbonneau (2004)

CLASS	WINKLER (°C·D)	GST (°C)
Too cold	< 850	< 13
Region I · Cold	850–1389	13–15
Region II · Intermediate	1390–1667	15–17
Region III · Temperate	1668–1944	17–19
Region IV · Warm	1945–2222	19–21
Region V · Very warm	2223–2700	21–24
Too hot	> 2700	> 24

Winkler and GST limits were checked against the legends of Figures 6 and 7 in the PDF of del Río et al. (2023). Winkler and GST classes are independent: the table puts them on the same row only by order. An index is computed if its period has at least 90 % of days with data (Ratio Vitis criterion); in the southern hemisphere periods shift by six months.

2.4 SPAC: dual-coefficient balance

$$ET_{c\ adj} = (K_s\ K_{cb} + K_e)\ ET_0$$

FAO-56, Eq. 80

$$K_s = (TAW - D_r) / ((1 - p)\ TAW) \text{ if } D_r > RAW \text{ ; otherwise } K_s = 1$$

FAO-56, Eq. 84

$$D_{r,i} = D_{r,i-1} - (P - RO)_i - I_i + ET_{c,i} + DP_i$$

FAO-56, Eq. 85

$$K_e = \min[K_r\ (K_{c,max} - K_{cb}),\ f_{ew}\ K_{c,max}]$$

FAO-56, Eq. 71

Initial, mid and late K_{cb}, p and height from FAO-56 Tables 17 and 22 for table or wine grapes; stages from Table 11; f_w from Table 20 by irrigation type. Flagged assumptions: RO = 0 and CR = 0; with drip, rain wets the whole surface (f_w = 1). Adjusting mid and late K_{cb} for wind and humidity (Eq. 70) is optional, and f_c can be entered or estimated with Eq. 76.

Ψ SCALE	CLASSES	SOURCE
Predawn leaf	≥ −0.3 · −0.3 a −0.5 · −0.5 a −0.8 · < −0.8 MPa	IVES; Carbonneau
Midday stem	≥ −0.6 · −0.7 a −1.1 · −1.2 a −1.6 · < −1.6 MPa	IVES; Deloire
Stem, irrigation (wine)	0 a −0.80 · −0.81 a −1.20 · −1.21 a −1.60 MPa	UC ANR
Stem, five levels	> −0.411 · −0.674 · −0.936 · −1.2 · < −1.2 MPa	Uriarte et al. (2025)

2.5 Fertility, biomass, productivity and energy

$$\mathrm{DOP} = C \cdot 100 / C_{ref} - 100 \; ; \; \Sigma |\mathrm{DOP}| = \text{overall imbalance}$$

Montañés et al. (1993); C_{ref}: Benito et al. (2015), Table 9

$$\text{dose (kg vine}^{-1}) = N_{obj} / [(N\,\% / 100)(1 - H\,\% / 100)] / 1000$$

$$LAI = LA / (d_h \, d_p) \; ; \; f_c = 0.98 \, (NDVI - 0.15) \; ; \; \Delta C = \Delta DM \cdot f_C \; ; \; CO_2 = \Delta C \cdot 44/12 \; ; \; O_2 = CO_2 \cdot 32/44$$

Er-Raki et al. (2013); Hernández González (2026)

$$1 - Y_a/Y_m = K_y \, (1 - ET_{c\,adj}/ET_c) \; ; \; K_y(\text{grape}) = 0.85$$

FAO-56, Eq. 90 and Table 24

Stress-free ET_c is rebuilt from the balance as $ET_{c\,adj} + \Sigma \, K_{cb}(1 - K_s) \, ET_0$, because K_e does not depend on K_s. Ravaz = yield / pruning (5–10) and leaf area/fruit ($0.8 \cdot 1.2 \cdot 1.6$ m² kg^{-1}) from UC Davis; table-grape harvest °Brix 14–17.5 (Permanhani et al., 2016). Leaf sufficiency ranges for red Grenache (Benito et al., 2015, Tables 5–8) and NC State for petiole at bloom.

$$LE = \lambda \, ET_c \; , \; \lambda = 2.45 \text{ MJ kg}^{-1} \; ; \; H = R_n - G - LE \; , \; G \approx 0 \text{ (daily)} \; ; \; \beta = H / LE$$

FAO-56, Eq. 42; OENO One

2.6 Statistics

$$y_{ij} = \mu + \tau_i + \beta_j + \gamma(x_{ij} - \bar{x}) + \varepsilon_{ij}$$

Block ANCOVA; Steel and Torrie (1980)

Descriptives, randomised-block ANOVA, Levene (median), Tukey with the studentized range distribution, Pearson, single-covariate ANCOVA and repeated measures as split plots in time (treatment against error a; date and treatment × date against error b). The proposal's mixed model (vine as random effect) is fitted outside, in R (lme4, emmeans), from the tidy CSV the block exports.

2.7 System network

$$C_B(v) = 2 / ((n-1)(n-2)) \cdot \Sigma_{s \neq v \neq t} \sigma_{st}(v) / \sigma_{st}$$

Brandes (2001)

Nodes are variables of blocks 1 to 9 and edges are signed directed relations from the network catalogue (`src/31_red.js`), each with its type and source. Simple directed cycles of length ≤ 7 are enumerated; a loop's sign is the product of its edge signs (negative: balancing; positive: reinforcing).

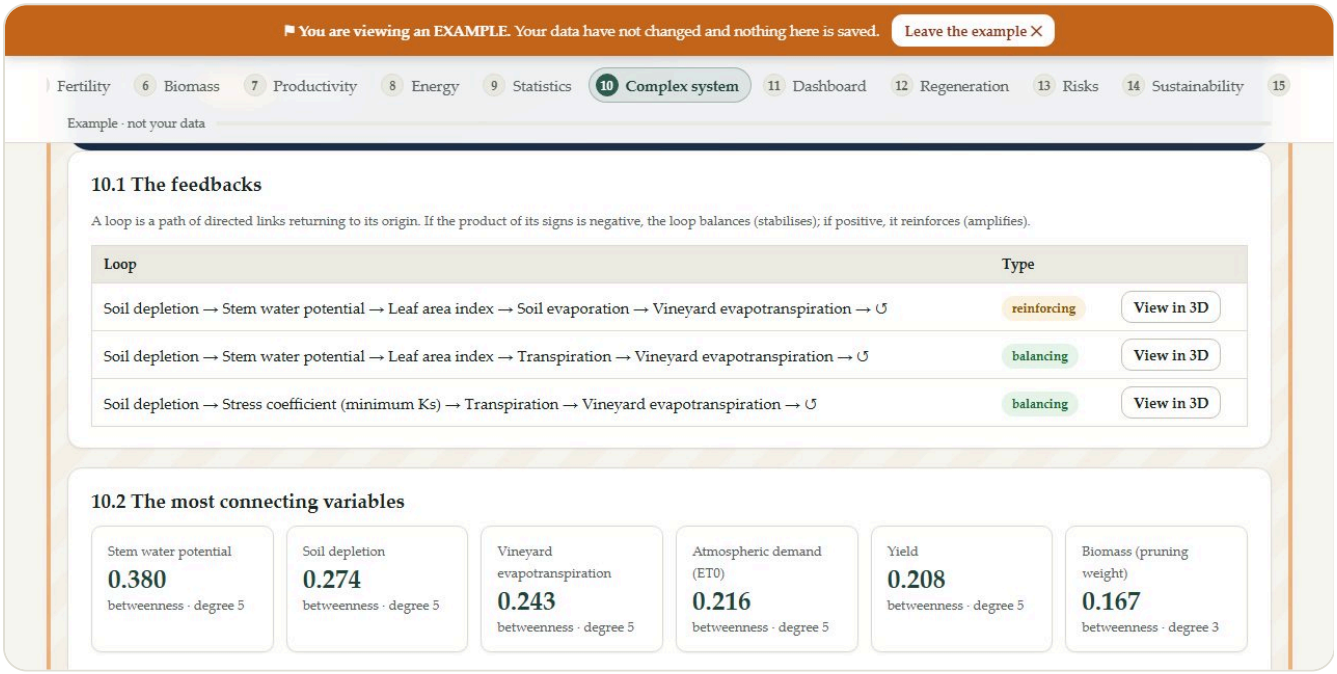


FIGURE 2.1 **Loops and bridge variables** with Mata de Zarza data.

2.8 Regeneration, risks and sustainability

$$IR = \Sigma s_k / (4\,n)\ ;\ r = P \cdot I \cdot (1 - 0.4\,CA) \cdot (1 - 0.15\,c)$$

Newton et al. (2020); IEC 31010:2019

$$I_n = (x - x_{min})/(x_{max} - x_{min})\ ;\ ID = \Sigma\, w_i\, I_{n,i} / \Sigma\, w_i\ ;\ ISG = \Sigma\, w_D\, ID / \Sigma\, w_D$$

Sarandón (2002); Hernández González (2026), 6.8

Sustainable: ISG ≥ 0.5 and every ID ≥ 0.5; 0–4 scale = ISG × 4. Footprint: N_2O = N (0.010 + 0.11·0.010 + 0.24·0.011) 44/28 with a GWP of 273 (IPCC, 2019; Forster et al., 2021), CO_2 from urea and fuel (IPCC, 2006). The risk coefficients 0.4 and 0.15 are flagged assumptions, the same as Ratio Viña.

2.9 Outlook

$$\theta_{1500t} = -0.024\,S + 0.487\,C + 0.006\,OM + 0.005\,S{\cdot}OM - 0.013\,C{\cdot}OM + 0.068\,S{\cdot}C + 0.031\ ;\ \theta_{1500} = \theta_{1500t} + (0.14\,\theta_{1500t} - 0.02)$$

Saxton and Rawls (2006), Eq. 1

$$\theta_{33t} = -0.251\,S + 0.195\,C + 0.011\,OM + 0.006\,S{\cdot}OM - 0.027\,C{\cdot}OM + 0.452\,S{\cdot}C + 0.299\ ;\ \theta_{33} = \theta_{33t} + (1.283\,\theta_{33t}^{2} - 0.374\,\theta_{33t} - 0.015)$$

Saxton and Rawls (2006), Eq. 2

$OM(t) = OM_0 + r{\cdot}t$ per treatment (OM capped at 8 %). Only $\Delta\theta = \theta(OM(t)) - \theta(OM_0)$ is added to the measured θ, with S and C from the jar test. Each scenario perturbs the base year by the delta method: $T + \Delta T$ (IPCC, 2018: 0.2 °C per decade; constant relative humidity, so T_{dew} shifts equally), rain $\times (1 + \Delta P)$ and irrigation $\times f_I$; the FAO-56 balance is run again. Resilience: Tr/Tpot, days with $K_s < 1$ and brake $\Sigma\, K_{cb} ET_0 (1 - K_s)$. The ISG is recomputed with a normalisation common to all years and scenarios. The hypothesis is tested with Spearman's ϱ between OM(5) and Tr/Tpot(5) per scenario.

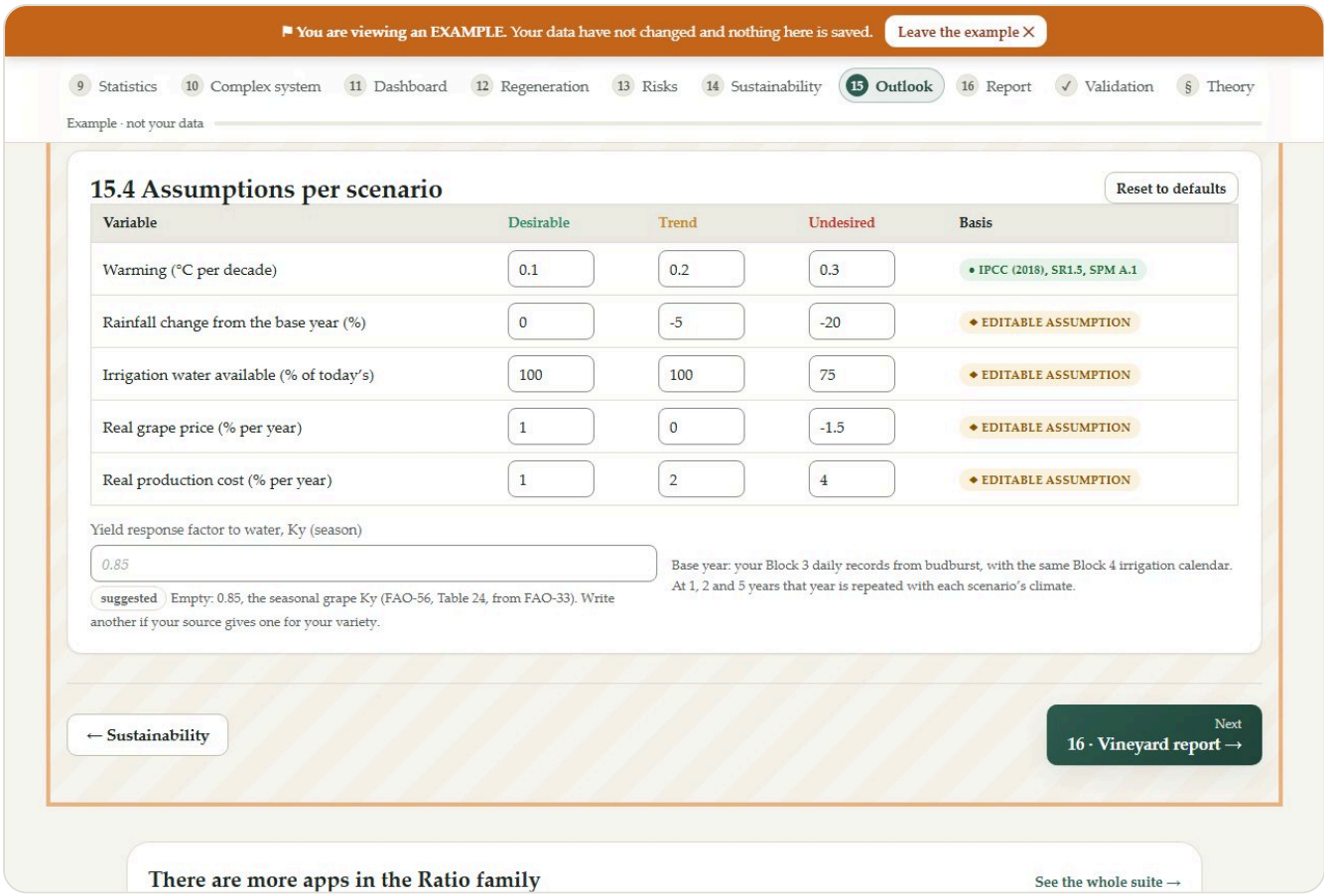


FIGURE 2.2 **Scenario assumptions.** Only ΔT is sourced (IPCC, 2018); the rest are flagged as editable.

THE CASE

3 Mata de Zarza, Veracruz

What data the example mode has, where they come from and what results.

Example mode loads a separate state that is never saved. It has three sources, and whatever has no measurement stays “no data”:

DATUM	VALUE	SOURCE
Site	Mata de Zarza, Puente Nacional, Ver.; 19.230° N, 467 m	INEGI
Grape, training, rootstock	Table; vertical trellis; own-rooted	Mojica-Zárate et al. (2026)
Spacing	3.0 × 1.40 m (2,381 vines ha^{-1})	Mojica-Zárate et al. (2026)
Planting; irrigation	February 2025 (approximate day: 24); drip	Mojica-Zárate et al. (2026)
Soil	Clay class; θ from Table 19 (suggested); Zr 0.70 m (◆)	Mojica-Zárate et al. (2026); FAO-56
Variety and design	Perlet; RCBD 4 × 3, 7-vine plots, 1 border; draw from the layout	Hernández González (2026)
Daily climate	2025, 365 days: Tmax, Tmin, RHmax, RHmin, Tdew, u10, Rs, P	ERA5 (Hersbach et al., 2020) via Open-Meteo (Zippenfeld, 2023)
Stages	Low latitudes, 20/40/120/60 d, from planting (◆)	FAO-56, Table 11

The climate is the ERA5 cell at 19.25° N, 96.75° W adjusted to 467 m, downloaded on 5 November 2026; 10 m wind is taken to 2 m with Eq. 47. Results the app gives:

OUTPUT	VALUE
Annual ET0 2025 (daily mean)	1,386 mm (3.80 mm d^{-1})
Rain 2025	1,765 mm
Winkler · GST	3,308 °C·d · 25.5 °C; both “too hot”
Huglin · CI	3,307 (HI+3) · 21.6 °C (CI–2)
TAW · RAW	98 mm · 34 mm
ETc adj from planting	1,085 mm: T 620 mm, E 465 mm (43 % evaporation)
Days with Ks < 1 · min Ks	125 of 311 · 0.22 (no irrigation recorded)
ETc adj/ETc · Ya/Ym (Ky 0.85)	0.89 · 0.90
Rn · LE · H · β	3,755 · 2,657 · 1,098 MJ m^{-2} · 0.41

! **LIMITS OF THE CASE**

With no irrigation recorded the balance treats the vine as rainfed, although the guide calls for constant irrigation during the first three months; and the K_{cb} are for mature vines while the plants are one to two years old. Both overstate 2025 stress and transpiration. They are corrected by entering irrigations and f_c or NDVI.

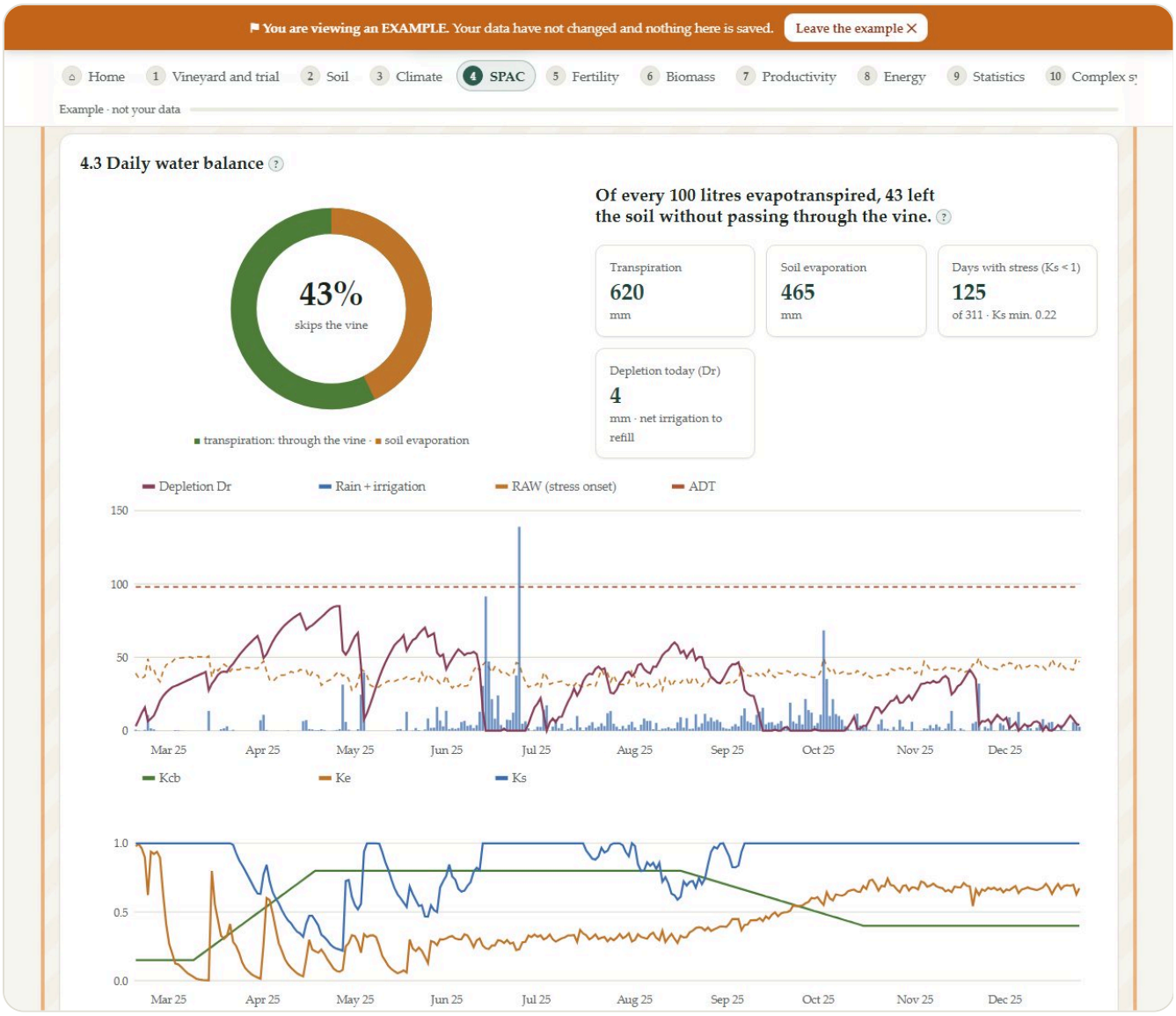


FIGURE 3.1 **2025 water balance in Mata de Zarza.**



FIGURE 3.2 **2025 energy partition.**

EVIDENCE AND USE

4 Validation and reproducibility

How the engine was checked and how to redo every calculation.

The app runs its tests every time you open the Validation block: **35 checks**, all correct in version 0.1.0.

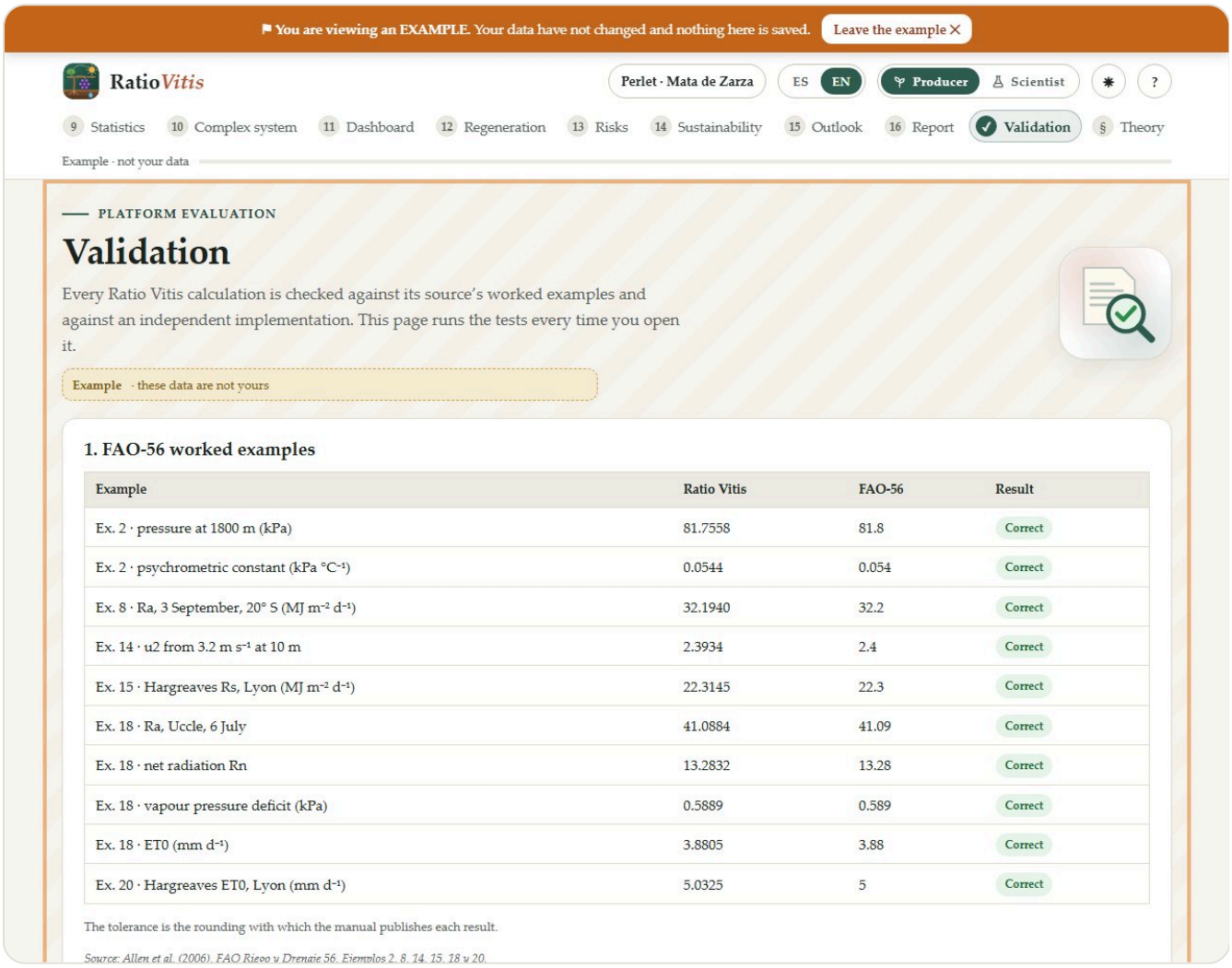


FIGURE 4.1 **Validation.** FAO-56 worked examples, independent engine, classes, statistics and network.

GROUP	WHAT IS COMPARED	RESULT
FAO-56	10 results of examples 2, 8, 14, 15, 18 and 20	within the published rounding
Independent engine	ET0 of 300 random days (14–33° latitude, 0–2600 m, with Rs, sunshine or Hargreaves) against `pruebas/verificar_fao56.py`	maximum difference 8.9 × 10^{-16} mm d^{-1}
Classes and formulas	USDA texture (5 cases), reproducible draw, Grenache, carbon, Er-Raki, IVES scale, DOP, Winkler/GST and Ky	exact
Statistics	ANOVA (40), Tukey, Levene, Pearson (30), ANCOVA (25), repeated measures (20) against SciPy and numpy (`pruebas/verificar_estadistica.py`)	≤ 10^{-8} in F, p and Tukey
Network	Degree, betweenness and cycles of 30 random graphs against NetworkX (`pruebas/verificar_red.py`)	difference ≤ 3 × 10^{-17}

◆ **ONE MORE TEST WITH THE CASE**

The app's annual ET0 for Mata de Zarza 2025 (1,385.6 mm, using T_{dew}) was recomputed with the independent Python engine and gave the same value.

4.1 Reproducing any number

In the scientist view, Report › Reproducible JSON downloads inputs (the full state), parameters, results and references; Block 3 exports climate in PhenologyPro format and Block 9 the tidy CSV for AgriDesign. The `pruebas/` scripts regenerate the test cases:

```
pruebas/

python verificar_fao56.py        # ejemplos de FAO-56 y 300 días → casos_fao56.json
python verificar_estadistica.py  # SciPy / numpy → casos_estadistica.json
python verificar_red.py          # NetworkX → casos_red.json
python ../armar.py               # incrusta los casos en index.html
```

4.2 Limitations

The balance uses a single soil layer and does not model capillary rise or runoff; viticultural indices were designed for temperate climates; leaf references are for red Grenache in La Rioja; Ky is seasonal, not per stage; the outlook uses exploratory scenarios, not forecasts, and the Saxton and Rawls pedotransfer was calibrated on US soils.

EVIDENCE AND USE

References

The 46 references the app uses, in APA 7.

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EVIDENCE AND USE

Glossary

Model and vineyard terms.

Dual crop coefficient *Coeficiente dual*	Splits ETc into transpiration (Ks·Kcb) and soil evaporation (Ke) (FAO-56, ch. 7).
Betweenness (Brandes) *Intermediación (Brandes)*	Share of shortest paths through a node, normalised by (n−1)(n−2).
Feedback loop *Lazo de retroalimentación*	Simple directed cycle in the network; its sign is the product of its edge signs.
Delta method *Método delta*	Climate scenario built by adding ΔT and scaling rainfall of the observed year.
mulberry32 *mulberry32*	Seeded 32-bit pseudo-random generator; makes the draw reproducible.
Min–max normalisation *Normalización min–máx*	Rescales an indicator to 0–1; reversed when less is better.
Pedotransfer function *Pedotransferencia*	Equation estimating water retention from sand, clay and organic matter (Saxton and Rawls, 2006).
Reanalysis *Reanálisis*	Climate reconstruction by a model with data assimilation (ERA5).
Relative transpiration *Transpiración relativa*	Tr/Tpot = 1 − ΣKcb·ET0(1−Ks)/ΣKcb·ET0; a water-resilience indicator.
Depletion (Dr) *Agotamiento (Dr)*	How much water the soil lacks, in millimetres, to return to field capacity. It is the net irrigation depth the app proposes.
RAW *AFA*	Readily available water: the part of available water the vine takes up without stress (p × TAW).
TAW *ADT*	Total available water in the root zone, between field capacity and wilting point, in mm.

AMIBA *AMIBA*	MESMIS radar chart comparing the sustainability indicators of each treatment.
ANOVA *ANOVA*	Analysis of variance: says whether the differences between treatments are larger than chance would give.
Block *Bloque*	In the trial, a group of similar plots (in Mata de Zarza, each row) where every treatment appears once.
Border vine *Bordo (planta de)*	A vine at the end of the row that receives the treatment but is not measured, to avoid edge effects.
Bowen ratio *Bowen (relación de)*	Sensible over latent heat. Low: the vineyard spends energy evaporating water; high: it heats the air.
Field capacity *Capacidad de campo*	The water the soil holds after excess has drained, one or two days after heavy irrigation.
Stress coefficient (Ks) *Coeficiente de estrés (Ks)*	Goes from 1 (no stress) to 0. Drops when depletion exceeds RAW (FAO-56).
DOP *DOP*	Deviation from optimum percentage of a nutrient: negative lacking, positive in excess (Montañés et al., 1993).
ET0 *ET0*	Reference evapotranspiration: how much water the air demands in a day, in mm, by FAO Penman–Monteith.
ETc *ETc*	Water the vineyard uses: vine transpiration plus soil evaporation.
GST *GST*	Growing season temperature, 1 April to 31 October in the northern hemisphere.
Huglin index *Índice de Huglin*	April–September heat sum that weighs sunny hours more (Tonietto and Carbonneau, 2004).
Ravaz index *Índice de Ravaz*	Kilos of fruit over kilos of pruning; between 5 and 10 the vine is balanced (UC Davis).
ISG *ISG*	General sustainability index, 0 to 1: sustainable at 0.5 or more with no dimension below (Sarandón, 2002).

Kcb *Kcb*	Basal crop coefficient: what share of ET0 the vine transpires in each stage (FAO-56, Table 17).
Ky *Ky*	Yield response factor to water; for grapes, 0.85 over the season (FAO-56, Table 24).
Ratio Vitis reading *Lectura de Ratio Vitis*	An app assumption to complete a calculation; always marked ◆ and editable.
Example mode *Modo ejemplo*	The Mata de Zarza case loaded separately: nothing is saved and your project does not change.
Plot *Parcela*	The experimental unit: a run of vines with the same treatment within a block.
Stem water potential *Potencial hídrico de tallo*	How much tension the water in the vine is under at midday, measured with a pressure chamber, in MPa.
Jar test *Prueba del frasco*	Shaking soil with water in a jar and measuring the sand, silt and clay layers as they settle.
Permanent wilting point *Punto de marchitez permanente*	The water left in the soil when the vine can no longer extract it.
SPAC *RASPA*	Soil–plant–atmosphere continuum: the path of water from rain and irrigation to the air, through the vine.
Reanalysis *Reanálisis*	Climate reconstructed from models and observations (such as ERA5); useful where there is no station, but not a measurement in your vineyard.
Feedback loop *Retroalimentación*	A network path that returns to its start: it balances (negative) or reinforces (positive).
Tukey *Tukey*	Test that compares treatments in pairs; different letters mean a real difference.
Winkler degree-days *Winkler (grados día)*	Heat useful to the vine from 1 April to 31 October, above 10 °C; defines regions I to V.

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How to cite the app

Mojica-Zárate, H. T., Hernández-González, J., Díaz-José, O., & Ramírez-García, A. G. (2026). *Ratio Vitis: The measured vineyard as a complex adaptive system: soil–plant–atmosphere water relations, fertility, biomass, productivity and energy* (Version 0.1.0) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.24048724

How to cite this guide

Mojica-Zárate, H. T., Hernández-González, J., Díaz-José, O., & Ramírez-García, A. G. (2026). *Ratio Vitis: Scientist Guide* (1st ed.). Zenodo. https://doi.org/10.5281/zenodo.24048729

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Imprint

Ratio Vitis · Scientist Guide · 1st edition, November 2026 · covers Ratio Vitis 0.1.0.

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ISBN: not assigned

DOI of this guide: 10.5281/zenodo.24048729 · DOI of the app: 10.5281/zenodo.24048724

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